

An inter-city optical clock network for precision metrology and relativistic geodesy

Haifeng Jiang

Abstract

A redefinition of the SI second relies on optical frequency ratios that different laboratories measure independently and that agree to within 5×10^{-18} . Many optical clocks report self-evaluated uncertainties meeting this level, but no comparison between clocks in different cities has yet reached it. Here we close this gap with an inter-city, cross-species clock network: an IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai, about 700 km apart, are linked through a 1350 km phase-stabilised fibre and compared coherently for 58 days (274 h, 16 segments with segment 9 excluded). Treating the tidal gravitational-redshift term as an uncertainty item, we obtain the Yb/Sr frequency ratio $1.207\,507\,039\,343\,337\,720\,1(29)$, with a statistical uncertainty of 0.88×10^{-18} and a total uncertainty of 2×10^{-18} , whose central value differs from the latest NIST–JILA result by less than 5×10^{-18} . The same data show a tidal-template response in the windowed beat series; fixed tidal-response scenarios change the inter-segment scatter descriptively but are not used to correct the headline ratio. The two clocks also read the time-varying gravitational potential over the real 700 km baseline, resolving a tidal geopotential difference at centimetre-level equivalent height rather than on a single campus. Such a network thus both supports the redefinition of the SI second and senses the time-varying gravitational potential.

1 Introduction

Optical clocks now measure frequency well enough to sense the gravitational redshift over a height difference of a centimetre¹, with self-evaluated fractional uncertainties below 10^{-18} ^{2,3} and two-clock stabilities approaching 10^{-18} within an hour⁴, and their accuracy is driving two goals at once: redefining the SI second on an optical transition and reading the Earth’s gravitational potential from clock comparisons^{5,6}. Both goals, however, depend on a capability that a single laboratory cannot provide. The best clocks are not transportable, so their claimed accuracy can only be confirmed by comparing them with one another over long distances. This has motivated the concept of a clock *network*, in which separate clocks are linked into a single cooperative instrument^{7,8}.

Fibre links have made such comparisons possible. Ultrastable light can now be transferred over hundreds to thousands of kilometres with added noise below the level of the clocks themselves^{9,10}, enabling inter-city links¹¹, multi-species ratio measurements¹², and, most recently, a fibre network spanning several countries¹³; transportable clocks, satellite links and free-space dissemination have extended comparisons further^{14–16}. Beyond metrology, such networks also probe fundamental physics, from gravitational waves¹⁷ to dark matter and topological defects¹⁸. Yet the comparisons reported so far fall short of what a redefined second requires. The CCTF roadmap calls for optical frequency ratios measured independently in different institutes and agreeing at the few-parts-in- 10^{18} level, whose closure also validates the estimated uncertainties^{5,19}: the only cross-species ratio reported at this level was measured over a short baseline between two clock laboratories in the same city²⁰, and the widest networks have not yet reached 5×10^{-18} ¹⁵.

It therefore remains unclear whether two clocks of different species, built at different institutions and separated by hundreds of kilometres of fibre, can agree at the level a redefined second

demands. Here we show that they can. We link an ytterbium lattice clock at IAPMST in Wuhan to a strontium lattice clock at USTC in Shanghai, about 700 km apart, through a 1350 km phase-stabilised fibre, and operate them coherently for 58 days. Treating the tidal gravitational-redshift term as an uncertainty item, we obtain a Yb/Sr ratio consistent with the existing determinations and observe a tidal-template response that requires covariance-aware follow-up before a formal detection claim.

2 Results

2.1 Network and observation campaign

The experiment compares a Yb lattice clock at IAPMST in Wuhan²¹ with a Sr lattice clock at USTC in Shanghai²², two institutions and two clock species about 700 km apart. They are joined by a phase-stabilised 1550 nm fibre link of single-pass optical length ≈ 1350 km, deployed on paired strands within a single cable duct of a telecommunications operator and organised as a cascade of relay stations, each carrying a bidirectional erbium-doped fibre amplifier and a fibre-noise-cancellation loop. Loop-back purification keeps the link instability below the clock noise beyond ~ 100 s (Fig. 1; Methods). The transfer architecture follows the cascaded, bias-corrected optical-frequency-dissemination design demonstrated on a 2067 km field link¹⁰; the present link is shorter, which relaxes the accumulated noise. The raw observable is the out-of-loop beat note between the received carrier and the local comb, sampled at 1 s.

Over a 58-day campaign (2026-06-29 to 2026-08-26), 16 uninterrupted segments (segment 9 excluded) yield 986 416 valid one-second samples (274 h, a duty cycle of about 20%), gated by the simultaneous lock of both clocks and the comb chain rather than by the link. The two clocks were restarted independently between segments in the first campaign, so the ratio centre provides a direct reproducibility check and shows no measurable bias across restarts. The static geopotential difference, $\Delta W = 280.042 \pm 0.085 \text{ m}^2/\text{s}^2$ (Wuhan higher), is fixed by first-order spirit levelling and applied as a constant correction (Methods); its uncertainty enters the ratio budget as the 1×10^{-18} static-redshift term (Table 1). For the metrological result we treat the time-varying tidal geopotential difference, $\Delta W(t) = W_{\text{WUHN}}(t) - W_{\text{SHAN}}(t)$, as an uncertainty item (Sec. 2.4), taking its waveform from a professional 30 s series that combines solid-Earth tide and ocean loading.

2.2 Clock-ratio determination

Combining the per-segment ratios with a random-effect model gives the uncorrected Yb/Sr frequency ratio

$$\text{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1(29), \quad (1)$$

with a Bayesian statistical uncertainty of 0.88×10^{-18} from the clock statistical noise alone. The parenthetical (29) is the display precision of the value string, not the statistical term. The alternative tidal-correction scenarios shift the centre by only a few 10^{-19} relative to the experiment-side tidal-corrected value, and an independent re-derivation from the raw beat files reproduces the per-segment ratios exactly (Methods).

Table 1 lists the uncertainty terms. The Yb-clock evaluation²¹ (1.1×10^{-18}) and the static gravitational redshift (1×10^{-18}) dominate; the Sr-clock term²² is 9.2×10^{-19} , the fibre and comb terms are each below 1×10^{-19} , and the tidal term is carried as an uncertainty item of $< 5 \times 10^{-19}$, the size of the theoretical compensation. Combined in quadrature these terms give a total fractional uncertainty of 2×10^{-18} . The clock-systematic terms are the published evaluations of the two clocks^{21;22}, and the tidal-model contribution is a theoretical bound; a full point-by-point systematics evaluation remains in progress.

Figure 2 shows the clock-ratio data behind Eq. (1): the 16 per-segment ratios R_i and the segment-to-segment stability. In terms of the deviations $y_i = R_i/R_{\text{seg1}} - 1$, the duration-weighted



Figure 1: Experimental setup. The IAPMST Yb lattice clock in Wuhan and the USTC Sr lattice clock in Shanghai are linked by a phase-stabilised 1550 nm fibre of single-pass optical length ≈ 1350 km, with loop-back purification at each repeater; the out-of-loop beat note between the received carrier and the local comb is the raw observable.

Table 1: Corrections and uncertainty terms for the Yb/Sr ratio (fractional frequency units). “Correction” is the applied shift to the central value; “Uncertainty” is the term’s contribution to the budget, combined in quadrature. The tidal term is carried as an uncertainty item (no correction applied; Sec. 2.4). The total is the quadrature sum of the listed terms.

Source	Correction	Uncertainty
Sr-clock systematic	9.2×10^{-19}	9.2×10^{-19}
Yb-clock systematic	1.1×10^{-18}	1.1×10^{-18}
Static gravitational redshift	-3.1×10^{-15}	1×10^{-18}
Fibre transfer	—	$< 1 \times 10^{-19}$
Comb transfer	—	$< 1 \times 10^{-19}$
Statistical (Bayesian)	—	0.88×10^{-18}
Tidal term (item)	—	$< 5 \times 10^{-19}$
Total	—	2×10^{-18}

mean is -0.639×10^{-18} (arithmetic mean -0.292×10^{-18}) and the segment-to-segment scatter is $\sigma(y_i) = 2.58 \times 10^{-18}$.

2.3 Tidal gravitational-redshift detection

The tide modulates the comparison frequency through the gravitational redshift, $\Delta f/f/f = \Delta W(t)/c^2$, with a segment-averaged rms amplitude of 3.90×10^{-18} (median; range 2.00 – 6.56×10^{-18} across the 16 segments).

After both the beat and a tidal template are passed through a 1200 s triangular window and demeaned, each segment is fitted with $A = \sum(\text{tide} \cdot \text{beat}) / \sum \text{tide}^2$. The template is normalised to the 1550 nm transfer frequency $F_{1550} = 193.40$ THz, not to the ratio-conversion coefficient $1/\text{COEF} \approx 233.53$ THz; the two differ by the Yb/Sr factor 1.2075, so the choice of reference sets the amplitude scale and must be accounted for before comparing A with other tidal analyses (Methods). A single segment reaches only $\text{SNR} \approx 0.28$. The 600-s-step windows overlap, so they cannot be used as independent samples for the historical p values or “sign-agnostic Stouffer” score.

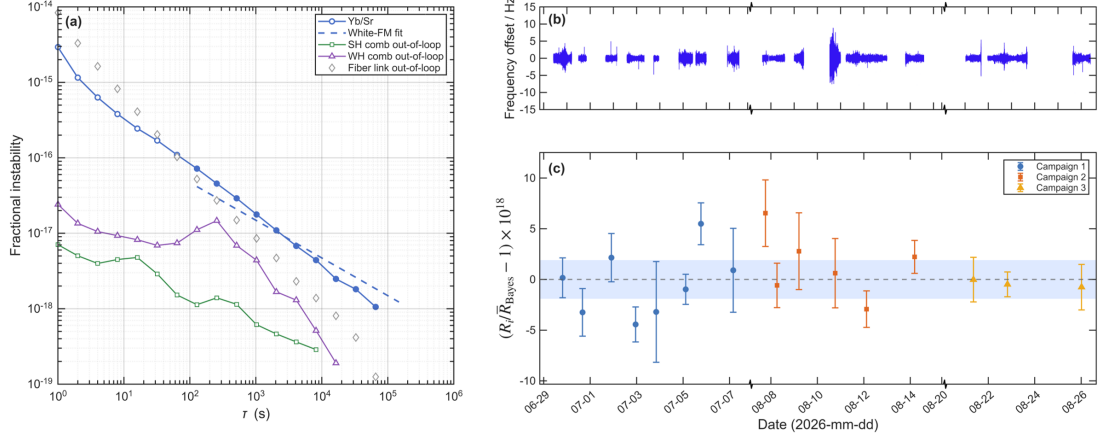


Figure 2: Clock-ratio data. The segment stability and the 16 per-segment Yb/Sr ratios R_i (segment 9 excluded) give the deviation $y_i = R_i/R_{\text{seg1}} - 1$ whose duration-weighted mean is -0.639×10^{-18} and whose segment-to-segment scatter is $\sigma(y_i) = 2.58 \times 10^{-18}$.

The amplitude point estimate uses all windows; its uncertainty and per-segment correlation are now evaluated from non-overlapping 1200-s windows, pending a full correlated-noise model. The historical fitted amplitude was approximately

$$A \approx -0.54, \quad (2)$$

and is consistent with a response of roughly half the theoretical magnitude. It is not used as a calibrated correction. At the segment-mean level, the ratio deviation y_i is positively correlated with the tidal shift $\Delta f/f$ (Pearson $r = +0.429$, $p = 0.098$).

2.4 Tidal effect and its compensation

At the full theoretical response ($A = -1$, complete trust in the tidal data and in general relativity) the tide would shift the ratio by 0.44×10^{-18} : in fractional-frequency terms the duration-weighted centre would move from 1.207 507 039 343 337 720 369 6 (raw) to 1.207 507 039 343 337 720 810 8. That is small against the 0.88×10^{-18} statistical term but comparable to the leading systematics, so we do not correct the central value; the tidal term enters the budget as an uncertainty item of $< 5 \times 10^{-19}$, its size set by that full-response shift (Table 1).

We asked whether an empirical response below the theoretical $A = -1$ is justified. The measured amplitude is $A = -0.54 \pm 0.08$ (Sec. 2.3), roughly half the theoretical magnitude, but no independent check establishes the fitted value as the correct response coefficient: the fit reproduces the waveform, not the hardware phase that fixes the sign convention. We therefore do not adopt it as a compensation.

A half-strength response admits several physical readings, none of which the present data can single out. The professional tidal series combines solid-Earth tide with ocean loading, yet the loading term over a 700 km inter-city baseline depends on the local coastline and sedimentary structure and on groundwater and hydrological loading that a global model resolves only coarsely; a phase- or amplitude-mismatched loading component would reduce the coherent response without removing it. Alternatively, the two clocks sense slightly different local geopotentials, and any common-mode environmental coupling along the fibre path that partially tracks the tide would also dilute the fitted amplitude. Because these mechanisms are degenerate at the present sensitivity, we keep the uncorrected centre as the headline ratio and report the amplitude as an effect estimate rather than a calibration, leaving a per-component tidal model as the natural route to a full-strength response.

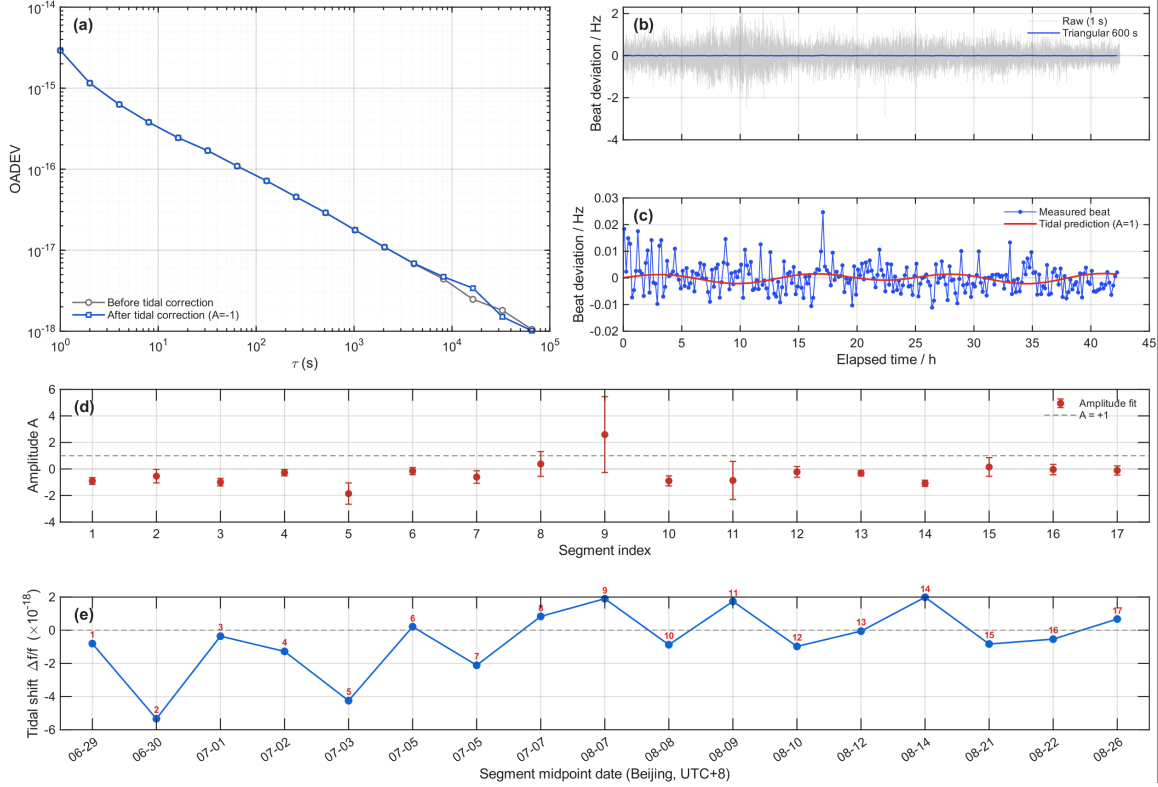


Figure 3: Tidal influence and its correlation. The 1200-s-window fit gives a historical amplitude estimate near $A = -0.54$ (about half the theoretical magnitude); overlapping-window significance is not interpreted as a detection statistic. At the segment-mean level the ratio deviation y_i is positively correlated with the tidal shift $\Delta f/f$ (Pearson $r = +0.429$).

Compensating at the fitted $A = -0.54$ would shift the duration-weighted centre by $+0.24 \times 10^{-18}$, half the $+0.44 \times 10^{-18}$ of the full theoretical response, both below the present statistical uncertainty. Descriptively, the empirical compensation would lower the reduced chi-square from 3.91 (raw) to 4.26 and the segment-to-segment dispersion $\sigma(y_i)$ from 2.580×10^{-18} to 2.334×10^{-18} (9.6%), while the full theoretical response gives a larger $\sigma(y_i)$ reduction to 2.473×10^{-18} (4.2%) but a slightly *higher* χ^2_{red} ($3.91 \rightarrow 4.11$); neither removes the residual at long averaging times. These are descriptive changes in scatter, not a validated correction: because neither the statistics nor the sign convention establishes it, the amplitude is reported as an effect estimate rather than adopted (Supplementary Information).

3 Discussion

Two features set this network apart. It is inter-city and cross-species: two clock species^{21,22}, at two institutions and two cities, joined only by the fibre link and the combs, yet yielding a ratio that agrees with the leading independent determinations at the 5×10^{-18} level without any transport of the clocks. And the same data set provides a time-varying-potential template response for further covariance-aware study; the 0.88×10^{-18} statistical floor corresponds to an equivalent height of about 0.8 cm.

The comparison places the Yb/Sr ratio on the map of independent determinations (Table 2, Fig. 4). Our uncorrected centre lies -2.9×10^{-18} below the NIST-JILA value²⁰ and $+2.1 \times 10^{-18}$ above the European-network value¹³. The NIST-JILA determination carries a total uncertainty at the few-parts-in- 10^{18} level, so our difference from it is below the 5×10^{-18} target; the European-network central value is quoted for completeness rather than as an independent agreement test,

Table 2: Yb/Sr frequency-ratio determinations. Deviations are given relative to the present work (uncorrected centre, Sec. 2.2), in units of 10^{-18} .

Determination	Yb/Sr	Deviation	Ref.
NIST–JILA	1.207 507 039 343 337 7230(37)	−2.9	²⁰
This work	1.207 507 039 343 337 720 1(29)	0	—
European network	1.207 507 039 343 337 718(32)	+2.1	¹³

since its statistical uncertainty is substantially larger, and the near-coincidence of the three values should not be read as a three-way agreement at 5×10^{-18} . The choice of uncorrected or tidal-corrected centre and of Bayesian or duration-weighted averaging moves any of these deviations by at most $\sim 0.7 \times 10^{-18}$, below the scale of interest.

The immediate metrological value of this work is as an independent check on the Yb/Sr ratio that the redefinition of the SI second will rest on. That redefinition calls for a matrix of ratios measured by different laboratories and mutually agreeing at the few-parts-in- 10^{18} level; for Yb/Sr only a single determination to date meets the 5×10^{-18} target²⁰, and it differs markedly from earlier reports. The present result adds a second, independently obtained entry to that matrix, from a third institution with a different pair of clocks and separate hardware and analysis, and so tests the redefinition on evidence the single-campus determination cannot provide. We stress that this is a conservative claim: the 0.88×10^{-18} figure is a statistical floor, and we report a ratio consistent with the existing determinations rather than a new most-precise value.

The same data set also bounds what the comparison can support. The bottleneck is no longer the link or the comb, whose contributions are each below 1×10^{-19} (Table 1), but the segment-to-segment scatter; closing the gap between the 0.88×10^{-18} statistical floor and the $\sim 5 \times 10^{-18}$ spread between independent determinations therefore requires progress on the systematic side, in the clock evaluations^{21;22} and in joint geodetic levelling, rather than longer integration alone. Two further limits should be read with the result: the per-segment uncertainties u_i are unreliable (an OADEV extrapolation that underestimates by roughly 30–40%, with anomalous segments 5, 6 and 16), so the combined values are for method comparison only; and the negative sign of the fitted amplitude is a specified response convention rather than a resolution of the hardware-polarity question.

4 Outlook

Beyond metrology, the same comparison is a candidate sensor for chronometric geodesy: the ratio and the geopotential difference are read from one measurement, and the tidal analysis here demonstrates the required sensitivity on a single baseline. A genuine relativistic coordinate reference frame, however, would require several nodes with closing loops, geodetic ties, a frame convention and continuous operation; a two-node, single-baseline, $\sim 20\%$ duty-cycle experiment is the elementary link of such a network, not the network itself. Extending this prototype node pair to a multi-node network, with improved link and comb stability and dedicated joint levelling, is the natural next step.

Beyond geodesy, the same network reaches into fundamental physics, where clock comparisons constrain dark-matter couplings^{23;24} and test Lorentz symmetry²⁵; realizing that reach on an inter-city baseline is an open question motivated by the present sensitivity.

5 Conclusion

We have demonstrated an inter-city, cross-species optical-clock network: an IAPMST Yb clock in Wuhan²¹ and a USTC Sr clock in Shanghai²², about 700 km apart and linked by a phase-

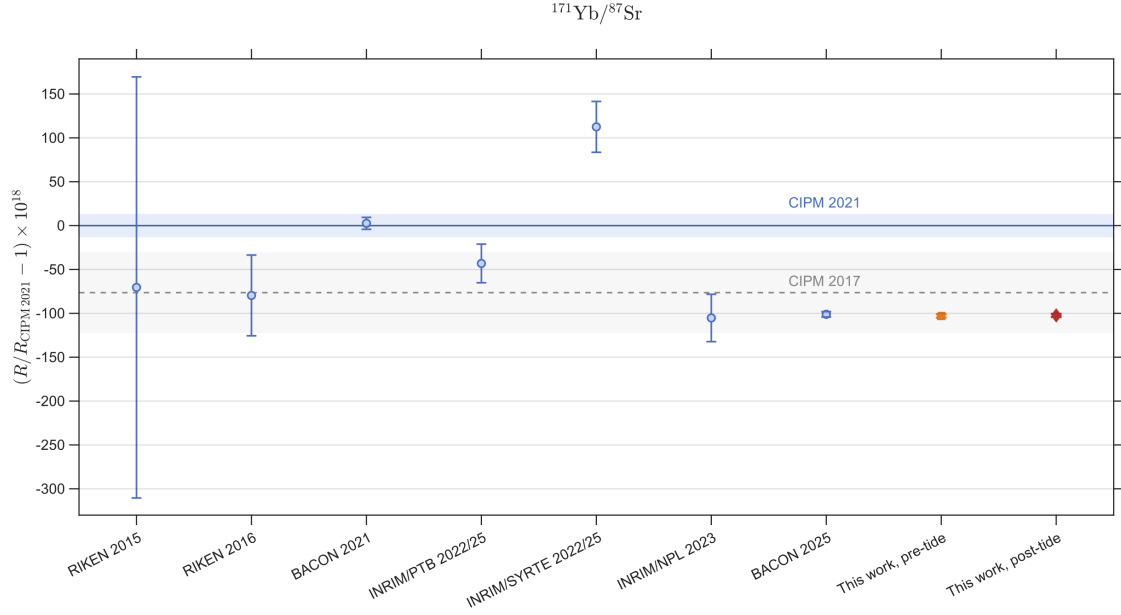


Figure 4: Comparison with previous determinations of the Yb/Sr ratio. The combined centres of this work, with the experiment-side WLS and NIST reference values, place the present result alongside the existing independent determinations.

stabilised fibre, operated coherently for 58 days. Combining 16 segments with a Bayesian random-effect model gives an uncorrected Yb/Sr ratio of $1.207\,507\,039\,343\,337\,720\,1(29)$ with a total uncertainty of 2×10^{-18} , consistent with the existing determinations at the 5×10^{-18} level. The tidal gravitational-redshift template response is carried as an uncertainty item, pending a covariance-aware detection analysis. Two capabilities follow. The comparison supplies an independently obtained entry for the Yb/Sr ratio that a redefined SI second will rest on. And, for the first time over a real 700 km baseline rather than a single campus, the two independent clocks resolve the time-varying gravitational potential at a centimetre-level equivalent height, casting the inter-city clock network as a prototype relativistic-geodesy sensor as well as a metrological one.

6 Methods

We describe the measurement and its analysis in three groups: the apparatus and the link (§“Experimental setup and frequency transfer”), the data set (§“Observation campaign and per-segment data”), and the analysis chain that turns the raw beat into the reported ratio—the ratio inversion, the systematic and geopotential corrections, the tidal treatment, and the statistical combination of the per-segment results.

6.1 Experimental setup and frequency transfer

The main text keeps only the conceptual frequency chain; here we record the symbol conventions that make the inversion reproducible. The observable is the beat between the received 1550 nm transfer carrier and the local Yb-referenced comb; every frequency in the chain is derived from the comb repetition rate $f_{\text{rep}} = f_{\text{ref}} \times \text{DIV}20 = 200 \text{ MHz}$.

The fibre link itself is built on the cascaded, bias-corrected optical-frequency-dissemination architecture of Ref.¹⁰: the route is divided into segments, each terminated by a relay station that carries a bidirectional erbium-doped fibre amplifier and a fibre-noise-cancellation loop, with

Table 3: AOM frequency offsets and their roles in the Shanghai–Wuhan link and comb chains (from the experiment-side supplementary material). FNC denotes a fibre-noise-cancellation loop; BBR a blackbody-radiation correction loop.

Index	Offset and role
1	+80 MHz, fixed
2	+80.274 MHz, FNC + closed-loop feedback
3	+80 MHz, FNC + BBR feedback
4	−110 MHz, FNC feedback
5	+50 MHz, FNC feedback
6	+236 MHz, clock-laser–cavity offset; cavity feed-forward
7	−80 MHz, FNC feedback
8	+155 MHz, frequency compensation and correction
9	−80 MHz, FNC feedback
10	+105 MHz, frequency compensation and correction
11	+80 MHz, FNC feedback
12	−140 MHz, clock interrogation control
a	+50 MHz, feedback, fibre transfer FNC
b	−40 MHz, fixed

the round-trip phase noise measured in the radio-frequency domain. Deployed on paired strands within a single telecommunications cable duct, the present Shanghai–Wuhan route is shorter than the 2067 km field link of Ref.¹⁰, so the accumulated link noise is correspondingly lower; the relay stations used here operate without the dedicated sub-hertz noise-purification stage that the longer link requires.

At the Shanghai end the Sr clock at 698 nm is transferred to a 1397 nm carrier; at the Wuhan end a second comb links the received 1550 nm carrier to the Yb clock at 578 nm through a 1156 nm fundamental beam. The clock-laser frequency relations are

$$f_{\text{comb}}^{1397} = \frac{1}{2} (1 + a) f_{\text{Sr}}, \quad a = a_{\text{dens}} + a_{\text{lat}} + a_{Z^2} + a_{\text{BGC}}, \quad (3)$$

$$f_{\text{comb}}^{1156} = \frac{1}{2} (1 + b) f_{\text{Yb}}, \quad (4)$$

where a collects the Sr-side systematic corrections²² and b the Yb-side correction²¹ ($b_{\text{Yb}} = -1.7 \times 10^{-18}$; the internal components are listed below and in the repository). Table 3 gives the AOM frequency offsets and their roles in the link and comb chains.

The static geopotential difference between the two clocks, $\Delta W = 280.04164 \text{ m}^2/\text{s}^2$, is applied at the ratio level as a constant correction $\delta_g = \Delta W/c^2 = -3.1159 \times 10^{-15}$ (fixed by the levelling described below): $(\text{Yb}/\text{Sr})_{\text{corr}} = (\text{Yb}/\text{Sr})_{\text{raw}} \times (1 + \delta_g)$.

6.2 Observation campaign and per-segment data

The comparison spans the 58-day campaign described in the main text. Table 4 lists the 16 uninterrupted segments used in this variant (segment 9 excluded), their retained sample counts n_i (after endpoint trimming) and measurement campaign; the retained total is $\sum_i n_i = 986\,416$ one-second samples. During campaign 1 the Yb and Sr clocks were restarted independently between some segments (Yb after segments 4–6; Sr after segments 7–8), which lets the reproducibility of the ratio be checked directly.

Table 4: The 16 uninterrupted segments (segment 9 excluded): index, retained samples n_i (endpoint-trimmed), and Beijing-time start.

Segment	n_i (s)	Start (Beijing)
1	68 009	2026-06-29 10:06:28
2	29 481	2026-06-30 12:00:01
3	42 891	2026-07-01 15:58:41
4	64 280	2026-07-02 14:00:01
5	19 532	2026-07-03 17:34:22
6	52 140	2026-07-04 19:30:46
7	39 526	2026-07-05 13:00:01
8	43 565	2026-07-06 21:45:16
10	57 938	2026-08-07 22:15:00
11	35 824	2026-08-09 00:00:00
12	40 346	2026-08-10 12:47:06
13	152 994	2026-08-11 05:30:00
14	68 618	2026-08-13 18:56:58
15	57 649	2026-08-21 00:40:03
16	147 640	2026-08-21 23:20:01
17	65 983	2026-08-25 15:09:41
Total	986 416	

6.3 Beat-to-ratio inversion

We take $b_i(t)$ as the longest valid beat sequence in segment i and m as the global median of the plausible beat values, so the segment beat deviation is

$$\overline{\delta b_i} = \overline{b_i} - m. \quad (5)$$

The Sr-side systematic shift a_i and the Yb-side shift b_{Yb} combine into the Dr coefficient, and the segment Yb/Sr ratio follows from the full inversion

$$R_i = [\text{ratio_base}_i + \text{Dr}_i]^{-1} \times (1 + \delta_g), \quad \delta_g = -3.1159 \times 10^{-15}, \quad (6)$$

where δ_g is the static gravitational correction. All ratio arithmetic is carried out at 80-decimal precision (`Decimal80`, mirroring `MATLAB vpa(...,80)`); a float64 absolute precision of about 2.6×10^{-16} would destroy differences at the 10^{-18} level. The duration-weighted centre of the whole experiment is

$$R_{\text{duration}} = \frac{\sum_i n_i R_i}{\sum_i n_i}, \quad \sum_i n_i = 986\,416. \quad (7)$$

6.4 Systematic frequency shifts

The ratio inversion removes the known systematic shifts of both clocks through the coefficients a_i (Sr side) and b (Yb side). The Yb-side correction is a fixed value, $b_{\text{Yb}} = (5.3 - 7.0) \times 10^{-18} = -1.7 \times 10^{-18}$, entering as $\text{coef}_{1156} = (1 + b_{\text{Yb}})/2$. The Sr-side correction is segment-dependent and is the sum of five terms,

$$a_i = a_{\text{rou},i} + a_{\text{AC},i} + a_{\text{SM},i} + a_{\text{air},i} + a_{\text{BBR},i}, \quad \text{coef}_{1397,i} = \frac{1}{2}(1 + a_i), \quad (8)$$

where a_{rou} is the lattice light-shift, a_{AC} the ac-Stark shift ($+7.444 \times 10^{-18}$ in the first campaign, $+8.929 \times 10^{-18}$ in the second), a_{SM} the second-order Zeeman shift ($\approx -1.78 \times 10^{-16}$

Table 5: Partial geopotential differences between the Yb (Wuhan) and Sr (Shanghai) clocks, in m^2/s^2 .

Segment	ΔW (m^2/s^2)
Yb atom $\rightarrow$ Yb optical platform	+2.65399
Yb platform $\rightarrow$ benchmark WS-03	−14.51715
Benchmark WS-03 $\rightarrow$ benchmark WS-01	+296.33022
Benchmark WS-01 $\rightarrow$ Sr optical platform	−1.47246
Sr platform $\rightarrow$ Sr atom	−2.95296
Yb atom $\rightarrow$ Sr atom (total)	+280.04164

in the first campaign; -8.925×10^{-17} in segment 9 and zero in segments 10–14), a_{air} the air-refractivity correction (-6.7×10^{-19}) and a_{BBR} the blackbody-radiation shift (zero). Segments 15–17 reuse the constant a_i of segments 12–14. The per-segment values are stored in the repository (`clock/params.json`); the anomalous `a_SM` of segment 9 is retained as delivered, without adjustment. The component evaluations follow the published assessments of the two clocks^{21;22}.

6.5 Levelling determination of the geopotential difference

The static gravitational-redshift correction δ_g entering the ratio inversion is fixed by the geopotential difference between the two clocks’ atomic transition positions, determined by first-order spirit levelling together with simultaneous gravimetry. A levelling line of about 1113 km was run between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC), comprising six first-order routes through Ezhou, Huangshi, Huanggang, Lu’an, Hefei, Wuhu, Guangde and Suzhou. Levelling was performed with digital levels (Sokkia SDL1X, Trimble DiNi03) and bar-coded invar staffs, with sight lengths below 30 m and a forward–back sight-length difference below 1.0 m; gravity was observed at every levelling point with two relative gravimeters. The levelled normal heights were converted to orthometric heights for the geopotential integration, and the geopotential difference between the two benchmarks follows from $\Delta W_{AB} = -\sum_i g_i \Delta H_i$, where g_i is the mean gravity and ΔH_i the orthometric height difference of the i -th segment.

Two further segments connect each benchmark to the corresponding atomic transition position: the benchmark-to-optical-platform height difference was obtained by trigonometrical heighting (total station), and the platform-to-atom segment from the clock design parameters. The five partial geopotential differences are listed in Table 5 and sum to $\Delta W = 280.04164 \text{ m}^2/\text{s}^2$ between the two atomic transition positions. The levelling precision, expressed as the accidental mean error per kilometre, is 0.255 mm, better than the 0.45 mm limit for first-order levelling. Propagating the levelling, trigonometrical-heighting and platform-to-atom terms yields a total height precision of $\pm 0.0086 \text{ m}$, and hence a geopotential-difference uncertainty of $\pm 0.085 \text{ m}^2/\text{s}^2$.

6.6 Tidal analysis

The tidal beat template is formed from the professional 30 s geopotential-difference series, linearly interpolated to the retained beat timestamps after shifting the Beijing-time beat windows to UTC. It is normalised to the 1550 nm transfer frequency, $h(t) = F_{1550} \Delta W(t)/c^2$ with $F_{1550} = N_{1550}^{\text{WH}} f_{\text{rep}} = 193.3992 \text{ THz}$, *not* to $1/\text{COEF}$, which would be larger by the Yb/Sr factor ≈ 1.2075 and would bias A low by that factor. Within each segment the beat and the template are passed through a 1200 s triangular window (600 s step) and, after demeaning, fitted with $\text{beat} = A \text{tide} + \text{noise}$, $A = \sum(\text{tide} \cdot \text{beat})/\sum \text{tide}^2$. The resulting segment-mean correlation between y_i and $\Delta f/f$ is shown in Fig. 5. We combine the per-segment amplitudes by precision weighting, together with the sign-consistency (binomial), signed Stouffer and Fisher summaries.

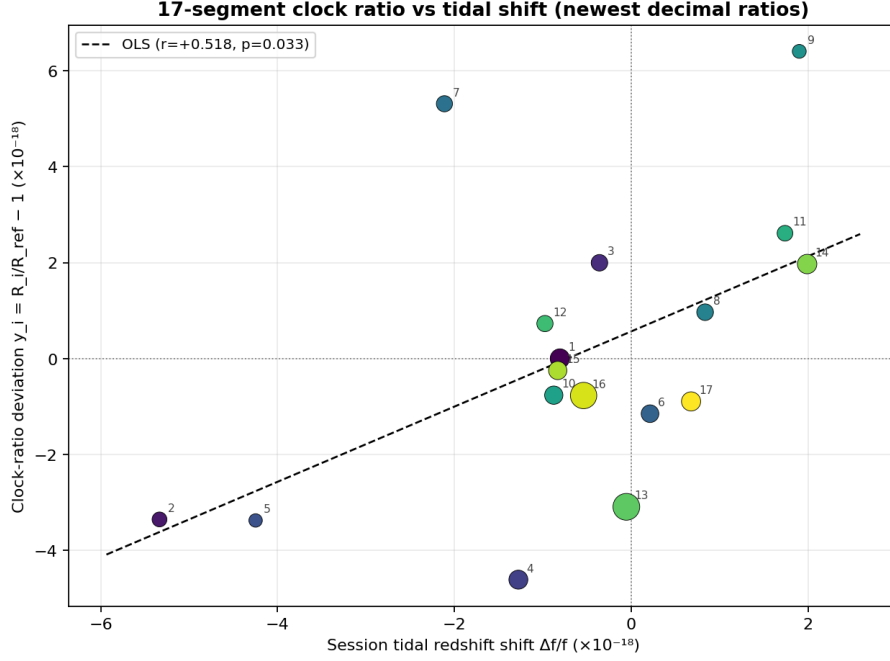


Figure 5: Segment-mean correlation. Clock-ratio deviation $y_i = R_i/R_{\text{seg1}} - 1$ against the session tidal shift $\Delta f/f$ for the 16 segments (segment 9 excluded); the ordinary least-squares fit gives the positive correlation reported in the Results (Pearson $r = +0.43$, $p = 0.098$).

The inferential inputs use non-overlapping windows; no sign-agnostic Stouffer significance is assigned.

For the fixed-coefficient scenarios we apply the correction to the beat before re-inverting the ratio, $b_{\text{corr}}(t) = b_{\text{raw}}(t) - A h(t)$ with $A = -1$ (theory) or $A = -0.54$ (empirical); the ratio chain is then re-inverted and the four combination methods re-applied. The direction self-check and the dispersion reduction are reported in the Results. Because the two amplitudes are fixed rather than fitted, and the sign follows a specified response convention (see Discussion), we report these scenarios as effect estimates and keep the uncorrected centre as the headline ratio.

6.7 Per-segment uncertainties

Each segment is assigned a statistical uncertainty from the overlapping Allan deviation (OADEV) of its fractional frequency series $\delta f/F_{1550}$. Over the white-frequency-noise range $128 \text{ s} \leq \tau \leq T/4$ the OADEV follows $\sigma_y(\tau) \propto \tau^{-1/2}$; a linear fit of $\log \sigma_y$ against $\log \tau$ is extrapolated to the segment length T to give $\sigma_y(T)$, and

$$u_i = \sigma_y(T) R_i \quad (9)$$

is the absolute segment uncertainty on R_i . A flicker floor beyond $\tau \sim 4096 \text{ s}$ makes this extrapolation underestimate the u_i (see Discussion). This is why we compare four combination methods instead of adopting a single u_i -weighted mean, and why the excess-scatter term ξ below is required rather than optional. The long-averaging-time behaviour is shown by the overlapping Allan deviation of the seventeen segments spliced end-to-end on one continuous 1 s record (gaps removed), for the raw, theory and empirical scenarios, with Riley–Howe effective-degrees-of-freedom error bars (Fig. 6).

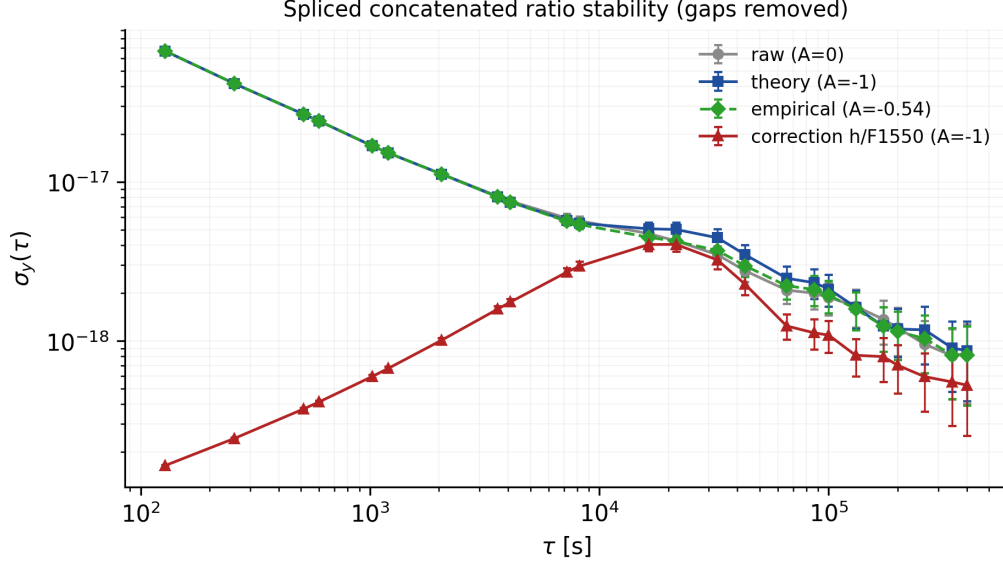


Figure 6: Spliced concatenated stability. Overlapping Allan deviation $\sigma_y(\tau)$ of the seventeen segments placed end-to-end on one continuous 1 s record (aligned pure-data span 11.7 d, gaps removed; τ up to 4.63 d), for the raw ($A = 0$), theory ($A = -1$) and empirical ($A = -0.54$) scenarios, with the correction-term curve and Riley–Howe EDF error bars.

6.8 Statistical combination methods

The inter-segment scatter exceeds what the per-segment uncertainties u_i alone can explain, so the final ratio is obtained with a random-effect model rather than a fixed-effect average. On the same per-segment (R_i, u_i) input we compare four combination methods, both for the uncorrected baseline and for the two tidal-correction scenarios (Sec. 6.6):

- **WLS (fixed effect)**: weights $w_i = 1/u_i^2$, no extra scatter; the u_i are taken at face value.
- **Birge**: the u_i are rescaled by a single factor $B = \sqrt{\chi_{\text{red}}^2}$.
- **Mandel–Paule**: a common excess random variance ξ^2 is added, $\sigma_i^2 = u_i^2 + \xi^2$, with ξ solved as a fixed value.
- **Bayesian hierarchical**: the same observation model as Mandel–Paule, but μ and ξ are both inferred, so the uncertainty of ξ propagates into the final ratio through the joint posterior (sampled by MCMC).

All four methods give consistent central values, and the statistical uncertainty grows monotonically from WLS through Birge and Mandel–Paule to Bayesian (Table 6); we adopt the Bayesian result as the primary statistical uncertainty. The direction of change under the tidal correction is the same for all four methods.

6.9 Statistical model details

Section 6.8 introduced the four combination methods. Here we record the shared observation model and the convergence diagnostics. All four methods assume

$$R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2), \quad (10)$$

where μ is the common true ratio and ξ the excess segment-to-segment scatter not described by the per-segment OADEV uncertainties u_i . The Bayesian hierarchical model treats both μ and ξ

Table 6: Centre and statistical uncertainty by combination method for the uncorrected baseline, computed in this work’s own re-run of the 16-segment ratios (segment 9 excluded), in units of 10^{-18} (deviations relative to the experiment-side WLS reference 1.207 507 039 343 337 721 3). The experiment-side analysis reports a slightly different Bayesian statistical term (0.88×10^{-18} ; adopted for this variant’s headline value), reflecting its own ratio pipeline.

Method	Centre deviation	Statistical u
WLS (fixed effect)	−1.152	0.286
Mandel–Paule	−0.661	0.785
Bayesian hierarchical	−0.656	0.879

as unknown and infers their joint posterior,

$$p(\mu, \xi \mid D) \propto p(D \mid \mu, \xi) p(\mu) p(\xi), \quad p(D \mid \mu, \xi) = \prod_i \frac{1}{\sqrt{2\pi(u_i^2 + \xi^2)}} e^{-\frac{(R_i - \mu)^2}{2(u_i^2 + \xi^2)}}, \quad (11)$$

with $D = \{R_i, u_i\}$. Because ξ is a random parameter, its posterior uncertainty propagates into the final ratio, unlike Mandel–Paule, which first fixes a point estimate of ξ and then averages.

The posterior is sampled by MCMC. Multiple independent chains mix over the full sampling range with no long-term drift or inter-chain separation, and the Gelman–Rubin convergence diagnostic, $\hat{R}_\mu = \hat{R}_\xi = 1.000$, confirms convergence to a common posterior. The final ratio is the posterior mean of μ , with the posterior standard deviation as the statistical uncertainty and the 68% credible interval taken from the posterior.

Funding

Author contributions

Competing interests

Acknowledgements

Data availability

The analysis code and derived numeric artifacts are available in the project repository at <https://github.com/hjiang555-a1ly/gravitation-whsh>, released under the Apache License 2.0. The raw beat files and the professional tidal series are not redistributed here; the derived per-segment ratios, the statistical-combination results, and the tidal-correction summaries are provided as machine-readable JSON and CSV files.

Code availability

The analysis code that produces the per-segment ratios, the statistical combinations, the tidal corrections and the leveling reduction is included in the same repository (<https://github.com/hjiang555-a1ly/gravitation-whsh>) under the Apache License 2.0. The raw beat files and the professional tidal series are not redistributed; the derived products are provided as machine-readable JSON and CSV files.

Supplementary information

Supplementary Information is provided with this manuscript, containing the per-segment data table (Supplementary Table 1), the full frequency-transfer chains and AOM offsets (Supplementary Fig. 1 and Supplementary Table 2), and the statistical-model details and MCMC convergence diagnostics (Supplementary Fig. 2).

References

- [1] W. F. McGrew, X. Zhang, R. J. Fasano, S. A. Schäffer, K. Beloy, D. Nicolodi, R. C. Brown, N. Hinkley, G. Milani, M. Schioppo, T. H. Yoon, and A. D. Ludlow. Atomic clock performance enabling geodesy below the centimetre level. *Nature*, 564:87–90, 2018. doi: 10.1038/s41586-018-0738-2. arXiv:1807.11282.
- [2] K. J. Arnold, M. D. K. Lee, Qi Zhao, Qichen Qin, Zhao Zhang, N. Jayjong, and M. D. Barrett. Lu^+ optical frequency references with accuracy verified at the 19th digit. *Nature*, 2026. doi: 10.1038/s41586-026-11072-8.
- [3] Bao-lin Zhang, Zi-xiao Ma, Yao Huang, Hui-li Han, Ru-ming Hu, Yu-zhuo Wang, Hua-qing Zhang, Li-yan Tang, Ting-yun Shi, Hua Guan, and Ke-lin Gao. Liquid-nitrogen-cooled $^{40}\text{Ca}^+$ ion optical clock with a systematic uncertainty of 4.4×10^{-19} . *Phys. Rev. Lett.*, 136: 053202, 2026. doi: 10.1103/vngc-c1xv.
- [4] E. Oelker, R. B. Hutson, C. J. Kennedy, L. Sonderhouse, T. Bothwell, A. Goban, D. Kedar, C. Sanner, J. M. Robinson, G. E. Marti, D. G. Matei, T. Legero, M. Giunta, R. Holzwarth, F. Riehle, U. Sterr, and J. Ye. Demonstration of 4.8×10^{-17} stability at 1 s for two independent optical clocks. *Nature Photonics*, 13(10):714–719, 2019. doi: 10.1038/s41566-019-0493-4.
- [5] N. Dimarcq, M. Gertszov, G. Milet, S. Bize, C. W. Oates, E. Peik, D. Calonico, T. Ido, P. Tavella, et al. Roadmap towards the redefinition of the second. *Metrologia*, 61(1):012001, 2024. doi: 10.1088/1681-7575/ad17d2. arXiv:2307.14141.
- [6] Jacopo Grotti, Silvio Koller, Stefan Vogt, Sebastian Häfner, Uwe Sterr, Christian Lisdat, Heiner Denker, Christian Voigt, Ludger Timmen, Antoine Rolland, Fred N. Baynes, Helen S. Margolis, Michał Zawada, Ursula Kleinevoss, Nils Huntemann, Ekkehard Peik, Marco Pizzocaro, Simone Donadello, Guglielmo M. Tino, Filippo Levi, Davide Calonico, Alberto Iaco, Marco Tarallo, Elio Rocco, Thomas König, Ali Al-Masoudi, Tanja Mehlstäubler, Christian Lück, Britta Borchert, and Harald Schnatz. Geodesy and metrology with a transportable optical clock. *Nature Physics*, 14:437–441, 2018. doi: 10.1038/s41567-017-0042-3.
- [7] Fritz Riehle. Optical clock networks. *Nature Photonics*, 11(1):25–31, 2017. doi: 10.1038/nphoton.2016.235.
- [8] P. Kómár, E. M. Kessler, M. Bishof, L. Jiang, A. S. Sørensen, J. Ye, and M. D. Lukin. A quantum network of clocks. *Nature Physics*, 10(8):582–587, 2014. doi: 10.1038/nphys3000.
- [9] Marco Schioppo, Jochen Kronjäger, Alissa Silva, Rada Ilieva, Chris W. Oates, Jacopo Tunesi, Charles F. A. Baynham, Helen S. Margolis, and Rachel M. Godun. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through 2220 km of optical fiber. *Nature Communications*, 13:212, 2022. doi: 10.1038/s41467-021-27884-3.
- [10] Fa-Xi Chen, Li-Bo Li, Jiu-Peng Chen, Kan Zhao, Jian-Yu Guan, Yang Xu, Lei Hou, Fei Zhou, Cheng-Zhi Peng, Qiang Zhang, Hai-Feng Jiang, and Jian-Wei Pan. 10^{-21} -level optical

- frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications*, 15:276, 2026. doi: 10.1038/s41377-026-02299-1.
- [11] C. Lisdat, G. Grosche, N. Quintin, C. Shi, S. M. F. Raupach, C. Grebing, D. Nicolodi, F. Stefani, A. Al-Masoudi, S. Dörscher, S. Häfner, et al. A clock network for geodesy and fundamental science. *Nature Communications*, 7:12443, 2016. doi: 10.1038/ncomms12443. arXiv:1511.07735.
 - [12] Kyle Beloy, Martha I. Bodine, Tobias Bothwell, Samuel M. Brewer, Sarah L. Bromley, Jwo-Sy Chen, Jean-Daniel Deschênes, Scott A. Diddams, Robert J. Fasano, Tara M. Fortier, Youssef S. Hassan, David B. Hume, Dhruv Kedar, Colin J. Kennedy, Isaac Khader, Amanda Koepke, David R. Leibbrandt, Holly Leopardi, Andrew D. Ludlow, William F. McGrew, William R. Milner, Nathan R. Newbury, Daniele Nicolodi, Eric Oelker, Thomas E. Parker, John M. Robinson, Stefania Romisch, Stefan A. Schäffer, Jeffrey A. Sherman, Laura C. Sinclair, Lindsay Sonderhouse, William C. Swann, Jian Yao, Jun Ye, and Xiaogang Zhang. Frequency ratio measurements at 18-digit accuracy using an optical clock network. *Nature*, 591:564–569, 2021. doi: 10.1038/s41586-021-03253-4. Boulder Atomic Clock Optical Network (BACON) Collaboration; arXiv:2005.14694.
 - [13] Marco Pizzocaro, Clara Zyskind, Anne Amy-Klein, Erik Benkler, Sebastien Bize, Davide Calonico, Etienne Cantin, Christian Chardonnet, Cecilia Clivati, Stefano Condio, E. Anne Curtis, Simone Donadello, Sören Dörscher, Chen-Hao Feng, Melina Filzinger, Jacques-Olivier Gaudron, Rachel M. Godun, Irene Goti, Ian R. Hill, Wei Huang, Nils Huntemann, Matthew Johnson, Joshua Klose, Jochen Kronjäger, Alexander Kuhl, Rodolphe Le Targat, Filippo Levi, Burghard Lipphardt, Christian Lisdat, Jerome Lodewyck, Olivier Lopez, Helen S. Margolis, Maxime Mazouth-Laurol, Alberto Mura, Benjamin Pointard, Paul-Eric Pottie, Matias Risaro, Billy I. Robertson, Marco Schioppo, Kilian Stahl, Martin Steinell, Alexandra Tofful, Mads Tø nnes, and Jacob Tunes. International optical clock comparison using the european optical fiber network. *Phys. Rev. Research*, 8:033250, 2026. doi: 10.1103/14bh-ryxs. arXiv:2604.27963.
 - [14] Ph. Laurent, F. X. Esnault, D. Massonnet, et al. The ACES/PHARAO space mission. *Comptes Rendus Physique*, 16(5):540–552, 2015. doi: 10.1016/j.crhy.2015.05.002.
 - [15] T. Lindvall, M. Pizzocaro, R. M. Godun, M. Abgrall, others, and H. S. Margolis. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica*, 12(6):843, 2025. doi: 10.1364/optica.561754.
 - [16] Qi Shen, Jian-Yu Guan, Ji-Gang Ren, Ting Zeng, Lei Hou, Meng Li, Yuan Cao, Jiu-Jiang Han, Meng-Yang Lian, Yin Chen, Xuan-Xuan Peng, Chao-Ze Wang, Deng-Ke Zou, Zhao Wang, Yu Wang, Gan-Sheng Pan, Juan Yin, Qiang Zhang, Cheng-Zhi Peng, and Jian-Wei Pan. Free-space dissemination of time and frequency with 10^{-19} instability over 113 km. *Nature*, 610:661–666, 2022. doi: 10.1038/s41586-022-05228-5.
 - [17] S. Kolkowitz, I. Pikovski, N. Langellier, M. D. Lukin, R. L. Walsworth, and J. Ye. Gravitational wave detection with optical lattice atomic clocks. *Physical Review D*, 94(12):124043, 2016. doi: 10.1103/PhysRevD.94.124043. arXiv:1606.01859.
 - [18] Andrei Derevianko and Maxim Pospelov. Hunting for topological dark matter with atomic clocks. *Nature Physics*, 10(12):933–936, 2014. doi: 10.1038/nphys3137.
 - [19] Fritz Riehle. Towards a redefinition of the second based on optical atomic clocks. *Comptes Rendus Physique*, 16(5):506–515, 2015. doi: 10.1016/j.crhy.2015.03.012. arXiv:1501.02068.

- [20] Alexander Aepli, Willa J. Arthur-Dworschack, Kyle Beloy, Caitlin M. Berry, Tobias Bothwell, Angela Folz, Tara M. Fortier, Tanner Grogan, Youssef S. Hassan, Zoey Z. Hu, David B. Hume, Benjamin D. Hunt, Kyungtae Kim, Amanda Koepke, Dahyeon Lee, David R. Leibbrandt, Ben Lewis, Andrew D. Ludlow, Mason C. Marshall, Nicholas V. Nardelli, Harikesh Ranganath, Daniel A. Rodriguez Castillo, Jeffrey A. Sherman, Jacob L. Siegel, Suzanne Thornton, William Warfield, and Jun Ye. Atomic clock frequency ratios with fractional uncertainty $\leq 3.2 \times 10^{-18}$. *Phys. Rev. Lett.*, 137:033201, 2026. doi: 10.1103/g865-9mk1. arXiv:2512.21428.
- [21] Qiang Zhu, Jingran Shi, Yuechen Zhang, Xiatian Xu, Bing Wang, Dezhi Xiong, Zhuanxian Xiong, Pengcheng Fang, Qunfeng Chen, Lingxiang He, and Baolong Lyu. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv preprint*, 2026. arXiv:2606.10514; submitted to Metrologia.
- [22] Zhi-Peng Jia, Jie Li, De-Quan Kong, Xiang Zhang, Hai-Wei Yu, Xiao-Yong Liu, Yu-Chen Zhang, Yuan-Bo Wang, Xian-Qing Zhu, Jia-Hao Zhang, Ming-Yi Zhu, Pei-Jun Feng, Xing-Yang Cui, Ping Xu, Xiao Jiang, Xiang-Pei Liu, Peng Liu, Han-Ning Dai, Yu-Ao Chen, and Jian-Wei Pan. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18} . *Metrologia*, 63:025002, 2026. doi: 10.1088/1681-7575/ae449e.
- [23] P. Wcisło, P. Ablewski, K. Beloy, S. Bilicki, M. Bober, R. Brown, R. Fasano, R. Ciurylo, H. Hachisu, K. Hosaka, et al. New bounds on dark matter coupling from a global network of optical atomic clocks. *Science Advances*, 4(12):eaau4869, 2018. doi: 10.1126/sciadv.aau4869.
- [24] B. M. Roberts, G. Blewitt, C. Dailey, M. Murphy, M. Pospelov, A. Rollings, J. Sherman, W. Williams, and A. Derevianko. Search for domain wall dark matter with atomic clocks on board global positioning system satellites. *Nature Communications*, 8:1195, 2017. doi: 10.1038/s41467-017-01440-4. arXiv:1704.06844.
- [25] Christian Sanner, Nils Huntemann, Richard Lange, Christian Tamm, Ekkehard Peik, Marianna S. Safronova, and Sergey G. Porsev. Optical clock comparison for lorentz symmetry testing. *Nature*, 567(7747):204–208, 2019. doi: 10.1038/s41586-019-0972-2.